BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Programming: Arrays and Iteration Practice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What makes a system embedded?

2. How many input combinations do three Boolean inputs have?

A) 3 B) 6
C) 9 D) 8

3. "Invalid test data must always have the wrong data type." Correct this. Explain why.

4. Draw a structure diagram: Quiz contains Menu, Play and Results. Play contains Input and Scoring. Why is this not a flowchart of running steps?

Today's objectives

Explain the core ideas of programming: arrays and iteration practice using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Array iteration
A loop visits elements by their valid indexes.

Accumulator
A running total starts at zero and adds each element.

Counting condition
A counter increments only when a tested item matches a rule.

Off-by-one error
Accessing index length or missing final element changes the result or causes a fault.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Array iteration	A loop visits elements by their valid indexes.
Accumulator	A running total starts at zero and adds each element.
Counting condition	A counter increments only when a tested item matches a rule.
Off-by-one error	Accessing index length or missing final element changes the result or causes a fault.

I Do - One worked example

Calculate the total of [3,5,2,4] using a running total.

1. Start total=0.
2. After values 3 and 5, total=8.
3. Add 2 to reach 10, then 4 to reach 14.
4. Print 14; the four valid indexes are 0-3.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Count values greater than 5 in [2,7,9,4].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the missing statement that adds the current file size.

```
array filesMB = [3,5,2]
total = 0
for i = 0 to 2
...
next i
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Give the output of summing [4,1,3]. [2 marks]

Q3. Explain why index 4 is invalid for a four-element zero-indexed array. [2 marks]

Q4. Explain how to count even values in an array. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A three-element zero-indexed array has last index 3. ☐ T ☐ F
2. A loop visits elements by their valid indexes. ☐ T ☐ F
3. A running total starts at zero and adds each element. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It is built into a larger device for a specific task, often with limited resources. ☐

Q2 D - 8 ☐

Q3 12 is an integer but outside the valid range, so it must be rejected. ☐

Q4 Quiz branches into Menu, Play and Results. Play branches into Input and Scoring. This shows parts and their hierarchy, rather than the order they run. ☐

Code First - Model

```
array filesMB = [3,5,2]
total = 0
for i = 0 to 2
total = total + filesMB[i]
next i
print(total)
```

Output: 10

We Do - Check the model

1. Final count=2. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 8. ☐

Q3 Valid indexes are 0,1,2,3; index 4 is out of bounds. ☐

Q4 Iterate each element; if value MOD 2 = 0 then increment counter; output counter. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: